CARDIA Y00, Y10, Y15, Y20

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2026-09-27

Reading Files: CARDIA Y00

```
# >>>>> YEAR 00 <<<<<<

library(readr)

#Be sure to replace the file path with the appropriate file path on your own computer.

base_dir <- "/Volumes/Data/Coronary Artery Risk Development in Young Adults (CARDIA)/Main_Study/Y00/DAT

file_paths <- c(
  file.path(base_dir, "aaf01.csv"),
  file.path(base_dir, "aaf08v2.csv"),
  file.path(base_dir, "aaf09med.csv"),
  file.path(base_dir, "aaf09gen.csv"),
  file.path(base_dir, "aachem.csv")
)

for (i in seq_along(file_paths)) {
  oname <- paste0("df", i)
  assign(oname, read_csv(file_paths[i]), envir = .GlobalEnv)
}
```

Cleaning: CARDIA Y00

```
#merging the data set
data_cardia_00 <- merge(df1, df2,  by ="RID", all = T)
data_cardia_00 <- merge(data_cardia_00, df3,  by ="RID", all = T)
data_cardia_00 <- merge(data_cardia_00, df4,  by ="RID", all = T)
data_cardia_00 <- merge(data_cardia_00, df5,  by ="RID", all = T)

# cleaning
rm(df1, df2, df3, df4, df5,
   df6, file_paths, i, oname)

# Cleaning data
attach(data_cardia_00)
```

```r
#creating new data frame which only has the variables of interest
df_cardia_00 <- data.frame(
  id_study = 'CARDIA_Y00',
  id       = as.character(RID),

  psu1     = NA,
  psu2     = NA,

  stratum1 = NA,
  stratum2 = NA,

  psu_weight1 = NA,
  psu_weight2 = NA,
  psu_weight3 = NA,

  # demographics
  sex     = A01SEX,
  age     = as.numeric(A01AGE2),
  race    = A01RACE2,
  edu_adult = NA, #level of education
  edu_years = as.numeric(A01ED2), #Years of education
  edu_child = NA,
  ethnicity = NA,

  #diabetes
  self_db      = A08DIAB, #diabetes ever
  age_told_db  = A09DIBAG,
  drug_insu_db = NA, #the question asked for for all types of med to control glucose, not "insulin" or
  drug_pill_db = NA, #the question asked for for all types of med to control glucose, not "insulin" or
  family_hx_db = NA,

  #labs
  fasting_minutes = NA,
  fasting_hours   = NA,
  fasting_8h      = NA,
  fasting_12h     = NA,
  fglu         = as.numeric(AL3_GLU), units_fglu  = 'mg/dL',
  hba1c        = NA, units_hba1c = 'NA',
  pogtt          = NA, units_pogtt   = 'NA',
  insulin      = NA, units_insulin = 'NA',

  creatinine_urine = NA, creatinine_urine_units = 'NA',

  total_cholesterol        = NA, units_cholesterol = 'mg/dL',
  ldl                      = NA, units_ldl = 'mg/dL',

  #tb
  tb_history = NA,
  age_told_tb  = NA,
  tb_treatment = NA,
  tb_skin_test = NA, #self-reported
  tb_skin_test_result = NA, #self-reported - ONLY FOR POSITIVE SKIN TEST
  tb_medication_use = NA,
```

```
  # Measurements
  weight                    = NA, units_weight = 'kg',
  height                    = NA, units_height = 'cm',
  bmi                       = NA, units_bmi = 'KG/M**2',
  waist                     = NA, units_waist = 'cm',            #mean of 2 measures
  hip                       = NA, units_hip = 'cm',             #mean of 2 measures
  blood_pressure_sys        = NA, units_sbp = 'mmHg',
  blood_pressure_diast      = NA, units_dbp = 'mmHg'
)

# Extract baseline demographics from Y00
baseline_demo <- df_cardia_00 %>%
  select(
    id,
    sex,
    age,
    race
  )


# Eliminating duplicates
df_cardia_00 <- df_cardia_00 %>%
  distinct(id, .keep_all = TRUE)


detach(data_cardia_00)
```

Reading Files: CARDIA Y10

```
# >>>>> YEAR 10 <<<<<<

library(readr)

#Be sure to replace the file path with the appropriate file path on your own computer.

base_dir <- "/Volumes/Data/Coronary Artery Risk Development in Young Adults (CARDIA)/Main_Study/Y10/DATA

file_paths <- c(
  file.path(base_dir, "earef.csv"),
  file.path(base_dir, "eaf08.csv"),
  file.path(base_dir, "eaf09tb.csv"),
  file.path(base_dir, "eaf09mhm.csv"),
  file.path(base_dir, "eaglu.csv"),
  file.path(base_dir, "eains.csv"),
  file.path(base_dir, "eachem.csv"),
  file.path(base_dir, "eaf02.csv"),
  file.path(base_dir, "eaf20.csv"),
  file.path(base_dir, "ealip.csv"),
  file.path(base_dir, "eaf05.csv")
)
```

```r
for (i in seq_along(file_paths)) {
  oname <- paste0("df", i)
  assign(oname, read_csv(file_paths[i]), envir = .GlobalEnv)
}
```

Cleaning: CARDIA Y10

```r
#merging the data set
data_cardia_10 <- merge(df1, df2,  by ="RID", all = T)
data_cardia_10 <- merge(data_cardia_10, df3,  by ="RID", all = T)
data_cardia_10 <- merge(data_cardia_10, df4,  by ="RID", all = T)
data_cardia_10 <- merge(data_cardia_10, df5,  by ="RID", all = T)
data_cardia_10 <- merge(data_cardia_10, df6,  by ="RID", all = T)
data_cardia_10 <- merge(data_cardia_10, df7,  by ="RID", all = T)
data_cardia_10 <- merge(data_cardia_10, df8,  by ="RID", all = T)
data_cardia_10 <- merge(data_cardia_10, df9,  by ="RID", all = T)
data_cardia_10 <- merge(data_cardia_10, df10,  by ="RID", all = T)
data_cardia_10 <- merge(data_cardia_10, df11,  by ="RID", all = T)



# cleaning
rm(df1, df2, df3, df4, df5,
   df6, df7, df8, df9, df10, df11, file_paths, i, oname)

# Cleaning data
attach(data_cardia_10)

#creating new data frame which only has the variables of interest
df_cardia_10 <- data.frame(
  id_study = 'CARDIA_Y10',
  id       = as.character(RID),

  psu1     = NA,       # check when merging
  psu2     = NA,

  stratum1 = NA,
  stratum2 = NA,

  psu_weight1 = NA,
  psu_weight2 = NA,
  psu_weight3 = NA,

  # demographics
  sex     = as.character(SEX),
  age     = as.numeric(EX5_AGE),
  race    = as.character(RACE),
  edu_adult = NA, #level of education
  edu_years = NA, #as.character(A03ED), #Years of education 10 = 10 or less years , 17 = 17 or more yea
  edu_child = NA,
  ethnicity = NA,
```

```
  #diabetes
  self_db      = as.character(E08DIAB), #diabetes ever
  age_told_db  = as.numeric(E08DIBAG),
  drug_insu_db = NA, #the question asked for for all types of med to control glucose, not "insulin" or
  drug_pill_db = NA, #the question asked for for all types of med to control glucose, not "insulin" or
  family_hx_db = NA,

  #labs
  fasting_minutes = as.numeric(E05FSTMN),
  fasting_hours   = NA,
  fasting_8h      = as.character(E05FST8H),
  fasting_12h     = as.character(E05FAST),
  fglu         = as.numeric(EL7GLU), units_fglu  = 'mg/dL',
  hba1c        = NA, units_hba1c = 'NA',
  pogtt          = as.numeric(EL7GLU2H), units_pogtt   = 'mg/dL',
  insulin      = as.numeric(EL7INS), units_insulin = 'uU/mL',

  creatinine_urine = as.numeric(EL7CREAT), creatinine_urine_units = 'mg/dL',

  total_cholesterol        = as.numeric(EL1CHOL), units_cholesterol = 'mg/dL',
  ldl                      = as.numeric(EL1LDL), units_ldl = 'mg/dL',

  #tb
  tb_history = as.character(E08TB),
  age_told_tb  = as.numeric(E08TBAG),
  tb_treatment = NA,
  tb_skin_test = NA, #self-reported
  tb_skin_test_result = as.character(E09TBPOS), #self-reported - ONLY FOR POSITIVE SKIN TEST
  tb_medication_use = NA,

  # Measurements
  weight                   = as.numeric(E20WGT), units_weight = 'kg',
  height                   = as.numeric(E20HGT), units_height = 'cm',
  bmi                      = as.numeric(E20BMI), units_bmi = 'KG/M**2',
  waist                    = as.numeric(E20WST), units_waist = 'cm',            #mean of 2 measures
  hip                      = as.numeric(E20HIP), units_hip = 'cm',            #mean of 2 measures
  blood_pressure_sys       = as.numeric(E02AVGSY), units_sbp = 'mmHg',
  blood_pressure_diast     = as.numeric(E02AVGDI), units_dbp = 'mmHg'
)

# Eliminating duplicates
df_cardia_10 <- df_cardia_10 %>%
  distinct(id, .keep_all = TRUE)

detach(data_cardia_10)
```

Reading Files: CARDIA Y15

```
# >>>>> YEAR 15 <<<<<<

library(readr)
```

```r
#Be sure to replace the file path with the appropriate file path on your own computer.

base_dir <- "/Volumes/Data/Coronary Artery Risk Development in Young Adults (CARDIA)/Main_Study/Y15/DAT

file_paths <- c(
  file.path(base_dir, "faref.csv"),
  file.path(base_dir, "faf03.csv"),
  file.path(base_dir, "faf08.csv"),
  file.path(base_dir, "faf09tb.csv"),
  file.path(base_dir, "faglu.csv"),
  file.path(base_dir, "fains.csv"),
  file.path(base_dir, "faf20.csv"),
  file.path(base_dir, "faf02.csv"),
  file.path(base_dir, "falip.csv"),
  file.path(base_dir, "faf05.csv")

)

for (i in seq_along(file_paths)) {
  oname <- paste0("df", i)
  assign(oname, read_csv(file_paths[i]), envir = .GlobalEnv)
}
```

Cleaning: CARDIA Y15

```r
#merging the data set
data_cardia_15 <- merge(df1, df2,  by ="RID", all = T)
data_cardia_15 <- merge(data_cardia_15, df3,  by ="RID", all = T)
data_cardia_15 <- merge(data_cardia_15, df4,  by ="RID", all = T)
data_cardia_15 <- merge(data_cardia_15, df5,  by ="RID", all = T)
data_cardia_15 <- merge(data_cardia_15, df6,  by ="RID", all = T)
data_cardia_15 <- merge(data_cardia_15, df7,  by ="RID", all = T)
data_cardia_15 <- merge(data_cardia_15, df8,  by ="RID", all = T)
data_cardia_15 <- merge(data_cardia_15, df9,  by ="RID", all = T)
data_cardia_15 <- merge(data_cardia_15, df10,  by ="RID", all = T)


# cleaning
rm(df1, df2, df3, df4, df5,
   df6, df7, df8, df9, df10, file_paths, i, oname)

attach(data_cardia_15)

#creating new data frame which only has the variables of interest
df_cardia_15 <- data.frame(
  id_study = 'CARDIA_Y15',
  id       = as.character(RID),

  psu1     = NA,       # check when merging
  psu2     = NA,

  stratum1 = NA,
```

```
stratum2 = NA,

psu_weight1 = NA,
psu_weight2 = NA,
psu_weight3 = NA,

# demographics
sex     = SEX,
#age    = will be borrowed from YOO (baseline)
race    = RACE,
edu_adult = F03DEGRE, #level of education
edu_years = as.numeric(F03ED.x),
edu_child = NA,
ethnicity = NA,

#diabetes
self_db      = F08DIAB, #diabetes ever
age_told_db  = F08DIBAG,
drug_insu_db = NA,
drug_pill_db = NA,
family_hx_db = NA,

#labs
fasting_minutes = as.numeric(F05FSTMN),
fasting_hours   = NA,
fasting_8h      = NA,
fasting_12h     = as.character(F05FAST),
fglu         = as.numeric(FL7GLU), units_fglu  = 'mg/dL',
hba1c        = NA, units_hba1c = 'NA',
pogtt          = NA, units_pogtt   = 'NA',
insulin      = as.numeric(FL7INS), units_insulin = 'uU/mL',

creatinine_urine = NA, creatinine_urine_units = 'NA',

total_cholesterol        = as.numeric(FL1CHOL), units_cholesterol = 'mg/dL',
ldl                      = as.numeric(FL1LDL), units_ldl = 'mg/dL',

#tb
tb_history = F08TB,
age_told_tb  = as.numeric(F08TBAG),
tb_treatment = NA,
tb_skin_test = NA,
tb_skin_test_result = F09TBPOS, #self-reported - ONLY FOR POSITIVE SKIN TEST
tb_medication_use = NA,

# Measurements
weight                   = as.numeric(F20WGT), units_weight = 'kg',
height                   = as.numeric(F20HGT), units_height = 'cm',
bmi                      = as.numeric(F20BMI), units_bmi = 'KG/M**2',
waist                    = as.numeric(F20WST1), units_waist = 'cm',
hip                      = NA,                  units_hip = 'cm',
blood_pressure_sys       = as.numeric(F02AVGSY), units_sbp = 'mmHg',
blood_pressure_diast     = as.numeric(F02AVGDI), units_dbp = 'mmHg'
```

```
)

detach(data_cardia_15)


### Borrowing "Age" from baseline (CARDIA - Y00)
df_cardia_15 <- df_cardia_15 %>%
  left_join(
    baseline_demo %>% select(id, age),
    by = "id"
  )

# Eliminating duplicates
df_cardia_15 <- df_cardia_15 %>%
  distinct(id, .keep_all = TRUE)
```

Reading Files: CARDIA Y20

```
# >>>>> YEAR 20 <<<<<<

library(readr)

#Be sure to replace the file path with the appropriate file path on your own computer.

base_dir <- "/Volumes/Data/Coronary Artery Risk Development in Young Adults (CARDIA)/Main_Study/Y20/DATA

file_paths <- c(
  file.path(base_dir, "gaf03.csv"),
  file.path(base_dir, "gaf08.csv"),
  file.path(base_dir, "gaf09tb.csv"),
  file.path(base_dir, "gaf09med.csv"),
  file.path(base_dir, "gaglu.csv"),
  file.path(base_dir, "gaf09med.csv"),
  file.path(base_dir, "gaf20.csv"),
  file.path(base_dir, "gaf02.csv"),
  file.path(base_dir, "galip.csv"),
  file.path(base_dir, "gaf05.csv")
)

for (i in seq_along(file_paths)) {
  oname <- paste0("df", i)
  assign(oname, read_csv(file_paths[i]), envir = .GlobalEnv)
}
```

Cleaning: CARDIA Y20

```
#merging the data set
data_cardia_20 <- merge(df1, df2,  by ="RID", all = T)
data_cardia_20 <- merge(data_cardia_20, df3,  by ="RID", all = T)
```

```r
data_cardia_20 <- merge(data_cardia_20, df4,  by ="RID", all = T)
data_cardia_20 <- merge(data_cardia_20, df5,  by ="RID", all = T)
data_cardia_20 <- merge(data_cardia_20, df6,  by ="RID", all = T)
data_cardia_20 <- merge(data_cardia_20, df7,  by ="RID", all = T)
data_cardia_20 <- merge(data_cardia_20, df8,  by ="RID", all = T)
data_cardia_20 <- merge(data_cardia_20, df9,  by ="RID", all = T)
data_cardia_20 <- merge(data_cardia_20, df10,  by ="RID", all = T)


# cleaning
rm(df1, df2, df3, df4, df5,
   df6, df7, df8, df9, df10, file_paths, i, oname)

attach(data_cardia_20)


#creating new data frame which only has the variables of interest
df_cardia_20 <- data.frame(
  id_study = 'CARDIA_Y20',
  id       = as.character(RID),

  psu1     = NA,       # check when merging
  psu2     = NA,

  stratum1 = NA,
  stratum2 = NA,

  psu_weight1 = NA,
  psu_weight2 = NA,
  psu_weight3 = NA,

  # demographics
  #sex     = will be borrowed from Y00 (baseline),
  #age     = will be borrowed from Y00 (baseline),
  #race    = will be borrowed from Y00 (baseline),
  edu_adult = G03DEGRE, #level of education
  edu_years = NA,
  edu_child = NA,
  ethnicity = NA,

  #diabetes
  self_db      = G08DIAB, #diabetes ever
  age_told_db  = G08DIBAG,
  drug_insu_db = NA, #G08MEDDIA: Are you taking medications for diabetes? 1=no / 2=yes / 8=not sure
  drug_pill_db = G08NOINS, #but only last 4 weeks, they asked for diet or pills
  family_hx_db = NA,

  #labs
  fasting_minutes = as.numeric(G05FSTMN),
  fasting_hours   = NA,
  fasting_8h      = NA,
  fasting_12h     = as.character(G05FAST),
  fglu         = as.numeric(GL7CGLU), units_fglu  = 'mg/dL',
```

```
  hba1c        = NA, units_hba1c = 'NA',
  pogtt          = as.numeric(GL7GLU2H), units_pogtt   = 'mg/dL',
  insulin      = NA, units_insulin = 'NA',

  creatinine_urine = NA, creatinine_urine_units = 'NA',

  total_cholesterol        = as.numeric(GL1CHOL), units_cholesterol = 'mg/dL',
  ldl                      = as.numeric(GL1LDL), units_ldl = 'mg/dL',

  #tb
  tb_history = G08TB,
  age_told_tb  = as.numeric(G08TBAG),
  tb_treatment = NA,
  tb_skin_test = G09TBPOS,
  tb_skin_test_result = NA, #F09TBPOS: self-reported - ONLY FOR POSITIVE SKIN TEST
  tb_medication_use = NA,

  # Measurements
  weight                   = as.numeric(G20WGT), units_weight = 'kg',
  height                   = as.numeric(G20HGT), units_height = 'cm',
  bmi                      = as.numeric(G20BMI), units_bmi = 'KG/M**2',
  waist                    = as.numeric(G20WST), units_waist = 'cm',
  hip                      = NA, units_hip = NA,
  blood_pressure_sys       = as.numeric(G02SAVG), units_sbp = 'mmHg',
  blood_pressure_diast     = as.numeric(G02DAVG), units_dbp = 'mmHg'
)

# Borrowing "Sex, Age, and Race" from Y00
df_cardia_20 <- df_cardia_20 %>%
  left_join(
    baseline_demo %>% select(id, sex, age, race),
    by = "id"
  )

detach(data_cardia_20)
```

Re-coding Y00, Y10, Y15, Y20 datasets

Cleaning numeric variables

```
# Summary of numeric variables - Y10
summary(df_cardia_10$weight)
summary(df_cardia_10$height)
summary(df_cardia_10$bmi)
summary(df_cardia_10$waist)
summary(df_cardia_10$hip)
summary(df_cardia_10$ldl)
summary(df_cardia_10$total_cholesterol)

# weight : lbs -> kg
```

```r
df_cardia_10 <- df_cardia_10 %>%
  mutate(weight = weight * 0.453592)


library(dplyr)

df_cardia_10 <- df_cardia_10 %>%
  mutate(
    weight = ifelse(weight >= 12 & weight <= 300, weight, NA),        # Plausible range: 12-300 kg => 2
    height = ifelse(height >= 100 & height <= 250, height, NA),             # Plausible range: 100 - 25
    bmi = ifelse(bmi >= 10 & bmi <= 80, bmi, NA),                           # Plausible range: 10-80 kg
    waist = ifelse(waist >= 30 & waist <= 300, waist, NA),                  # Plausible range: 30 - 300
    hip = ifelse(hip >= 40 & hip <= 300, hip, NA),                          # Plausible range: 40 - 300
    ldl = ifelse(ldl >= 19.33 & ldl <= 386.65, ldl, NA),                    # Plausible range: 0.5 - 10
    total_cholesterol = ifelse(
      total_cholesterol >= 1.75 & total_cholesterol <= 20,                  # Plausible range: 1.75 - 2
      total_cholesterol, NA
    )
  )

# Summary of numeric variables - Y15
summary(df_cardia_15$weight)
summary(df_cardia_15$height)
summary(df_cardia_15$bmi)
summary(df_cardia_15$waist)
summary(df_cardia_15$hip)
summary(df_cardia_15$ldl)
summary(df_cardia_15$total_cholesterol)

# weight : lbs -> kg
df_cardia_15 <- df_cardia_15 %>%
  mutate(weight = weight * 0.453592)

df_cardia_15 <- df_cardia_15 %>%
  mutate(
    weight = ifelse(weight >= 12 & weight <= 300, weight, NA),        # Plausible range: 12-300 kg => 2
    height = ifelse(height >= 100 & height <= 250, height, NA),             # Plausible range: 100 - 25
    bmi = ifelse(bmi >= 10 & bmi <= 80, bmi, NA),                           # Plausible range: 10-80 kg
    waist = ifelse(waist >= 30 & waist <= 300, waist, NA),                  # Plausible range: 30 - 300
    hip = ifelse(hip >= 40 & hip <= 300, hip, NA),                          # Plausible range: 40 - 300
    ldl = ifelse(ldl >= 19.33 & ldl <= 386.65, ldl, NA),                    # Plausible range: 0.5 - 10
    total_cholesterol = ifelse(
      total_cholesterol >= 1.75 & total_cholesterol <= 20,                  # Plausible range: 1.75 - 2
      total_cholesterol, NA
    )
  )

# Summary of numeric variables - Y20
summary(df_cardia_20$weight)
summary(df_cardia_20$height)
summary(df_cardia_20$bmi)
summary(df_cardia_20$waist)
summary(df_cardia_20$hip)
```

```r
summary(df_cardia_20$ldl)
summary(df_cardia_20$total_cholesterol)

# weight : lbs -> kg
df_cardia_20 <- df_cardia_20 %>%
  mutate(weight = weight * 0.453592)


df_cardia_20 <- df_cardia_20 %>%
  mutate(
    weight = ifelse(weight >= 12 & weight <= 300, weight, NA),        # Plausible range: 12-300 kg => 2
    height = ifelse(height >= 100 & height <= 250, height, NA),             # Plausible range: 100 - 25
    bmi = ifelse(bmi >= 10 & bmi <= 80, bmi, NA),                           # Plausible range: 10-80 kg
    waist = ifelse(waist >= 30 & waist <= 300, waist, NA),                  # Plausible range: 30 - 300
    hip = ifelse(hip >= 40 & hip <= 300, hip, NA),                          # Plausible range: 40 - 300
    ldl = ifelse(ldl >= 19.33 & ldl <= 386.65, ldl, NA),                    # Plausible range: 0.5 - 10
    total_cholesterol = ifelse(
      total_cholesterol >= 1.75 & total_cholesterol <= 20,                  # Plausible range: 1.75 - 2
      total_cholesterol, NA
    )
  )
```

Blood pressure

```r
# Summary of bp - Y10
summary(df_cardia_10$blood_pressure_sys)
summary(df_cardia_10$blood_pressure_diast)

library(dplyr)
df_cardia_10 <- df_cardia_10 %>%
  mutate(
    blood_pressure_sys = ifelse(
      blood_pressure_sys >= 70 & blood_pressure_sys <= 270,
      blood_pressure_sys, NA
    ),
    blood_pressure_diast = ifelse(
      blood_pressure_diast >= 30 & blood_pressure_diast <= 150,
      blood_pressure_diast, NA
    )
  )

# Summary of bp - Y15
summary(df_cardia_15$blood_pressure_sys)
summary(df_cardia_15$blood_pressure_diast)

df_cardia_15 <- df_cardia_15 %>%
  mutate(
    blood_pressure_sys = ifelse(
      blood_pressure_sys >= 70 & blood_pressure_sys <= 270,
      blood_pressure_sys, NA
    ),
```

```
    blood_pressure_diast = ifelse(
      blood_pressure_diast >= 30 & blood_pressure_diast <= 150,
      blood_pressure_diast, NA
    )
  )

# Summary of bp - Y20
summary(df_cardia_20$blood_pressure_sys)
summary(df_cardia_20$blood_pressure_diast)

df_cardia_20 <- df_cardia_20 %>%
  mutate(
    blood_pressure_sys = ifelse(
      blood_pressure_sys >= 70 & blood_pressure_sys <= 270,
      blood_pressure_sys, NA
    ),
    blood_pressure_diast = ifelse(
      blood_pressure_diast >= 30 & blood_pressure_diast <= 150,
      blood_pressure_diast, NA
    )
  )
```

Diabetes definitions

```
# Transforming minutes to hours -> 8h cut off
df_cardia_15 <- df_cardia_15 %>%
  mutate(
    fasting_8h = case_when(
      !is.na(fasting_minutes) & fasting_minutes >= 480 ~ 1,    #  8 hours (480 min)
      !is.na(fasting_minutes) & fasting_minutes < 480 ~ 0,     # < 8 hours
      TRUE ~ NA_real_
    )
  )

df_cardia_20 <- df_cardia_20 %>%
  mutate(
    fasting_8h = case_when(
      !is.na(fasting_minutes) & fasting_minutes >= 480 ~ 1,    #  8 hours (480 min)
      !is.na(fasting_minutes) & fasting_minutes < 480 ~ 0,     # < 8 hours
      TRUE ~ NA_real_
    )
  )

# Cleaning fasting glucose:
df_cardia_10 <- df_cardia_10 %>%
  mutate(
    fglu = case_when(
      fasting_8h == 0 ~ NA_real_,      # NA increased: 1189 -> 1268
      TRUE ~ fglu
    )
  )
```

```r
df_cardia_15 <- df_cardia_15 %>%
  mutate(
    fglu = case_when(
      fasting_12h == 0 ~ NA_real_,    # NA increased: 1409 -> 1672
      TRUE ~ fglu
    )
  )


df_cardia_20 <- df_cardia_20 %>%
  mutate(
    fglu = case_when(
      fasting_12h == 0 ~ NA_real_,     # NA increased: 58 -> 280
      TRUE ~ fglu
    )
  )


# Diabetes: Self reported / labs (fasting glucose, HbA1c, oral glucose tolerance test)
create_self_lab_dm <- function(dm) {
  dm %>%
    mutate(
      self_lab_dm = case_when(
        self_db == 1 | fglu >= 126 | hba1c >= 6.5 | pogtt >= 200 ~ 1,
        !is.na(self_db) | !is.na(fglu) | !is.na(hba1c) | !is.na(pogtt) ~ 0,
        is.na(self_db) & is.na(fglu) & is.na(hba1c) & is.na(pogtt) ~ NA_real_
      )
    )
}

df_cardia_00 <- create_self_lab_dm(df_cardia_00)
df_cardia_10 <- create_self_lab_dm(df_cardia_10)
df_cardia_15 <- create_self_lab_dm(df_cardia_15)
df_cardia_20 <- create_self_lab_dm(df_cardia_20)
```

Labeling both datasets

```r
label_cardia <- function(label) {
  label %>%
    mutate(
      # Sex
      sex = factor(sex,
                   levels = c(1, 2),
                   labels = c("Men", "Women")),

      # TB variables
      tb_history = factor(tb_history,
                          levels = c(1, 0, 9),
                          labels = c("Yes", "No", "Refused/DK")),
```

```
    tb_treatment = factor(tb_treatment,
                          levels = c(1, 0, 9),
                          labels = c("Yes", "No", "Refused/DK")),
    tb_skin_test = factor(tb_skin_test,
                          levels = c(1, 0, 9),
                          labels = c("Yes", "No", "Refused/DK")),
    tb_skin_test_result = factor(tb_skin_test_result,
                                 levels = c(1, 0, 9),
                                 labels = c("Positive", "Negative", "Refused/DK")),
    tb_medication_use = factor(tb_medication_use,
                               levels = c(1, 0, 9),
                               labels = c("Yes", "No", "Refused/DK")),

    # Diabetes self-report + labs
    self_db = factor(self_db,
                     levels = c(1, 0, 9),
                     labels = c("Yes", "No", "Refused/DK")),
    drug_insu_db = factor(drug_insu_db,
                          levels = c(1, 0, 9),
                          labels = c("Yes", "No", "Refused/DK")),
    drug_pill_db = factor(drug_pill_db,
                          levels = c(1, 0, 9),
                          labels = c("Yes", "No", "Refused/DK")),
    self_lab_dm = factor(self_lab_dm,
                         levels = c(1, 0),
                         labels = c("Diabetes", "No Diabetes")),

    # Family history
    family_hx_db = factor(family_hx_db,
                          levels = c(1, 0),
                          labels = c("Yes", "No")),

    # Race/Ethnicity
    race = factor(race,
                  levels = c(0, 1, 2, 3, 9),
                  labels = c("Non-Hispanic Black",
                             "Non-Hispanic White",
                             "Mexican/Other Hispanic",
                             "Other/Multiracial",
                             "Missing")),
    ethnicity = factor(ethnicity,
                       levels = c(1, 2, 9),
                       labels = c("Non-Hispanic", "Hispanic", "Missing")),

    # Education
    edu_adult = factor(edu_adult,
                       levels = c(1, 2, 3, 4, 9),
                       labels = c("< High School",
                                  "High School Grad",
                                  "Some College/AA",
                                  "College+",
                                  "Refused/DK"))
)
```

```
}

df_cardia_00 <- label_cardia(df_cardia_00)
df_cardia_10 <- label_cardia(df_cardia_10)
df_cardia_15 <- label_cardia(df_cardia_15)
df_cardia_20 <- label_cardia(df_cardia_20)
```

Exploratory data analysis

```
###==== Year 00 =====
#TB (n=0)
table(df_cardia_00$tb_history)

#DM
table(df_cardia_00$self_db) #(n=42)
table(df_cardia_00$self_lab_dm) #(n=55)

#DM and TB (n=0)
table(df_cardia_00$tb_history, df_cardia_00$self_db)
table(df_cardia_00$tb_history, df_cardia_00$self_lab_dm)

###==== Year 10 =====
#TB (n=80)
table(df_cardia_10$tb_history)

#DM
table(df_cardia_10$self_db) # (n=156)
table(df_cardia_10$self_lab_dm) #(n=187)

#DM and TB
table(df_cardia_10$tb_history, df_cardia_10$self_db) # (n=4)
table(df_cardia_10$tb_history, df_cardia_10$self_lab_dm) # (n=4)

###==== Year 15 =====
#TB (n=112)
table(df_cardia_15$tb_history)

#DM
table(df_cardia_15$self_db) # (n=190)
table(df_cardia_15$self_lab_dm) #(n=213)


#DM and TB
table(df_cardia_15$tb_history, df_cardia_15$self_db) # (n=2)
table(df_cardia_15$tb_history, df_cardia_15$self_lab_dm) # (n=2)

###==== Year 20 =====
#TB (n=125)
table(df_cardia_20$tb_history)

#DM
```

```r
table(df_cardia_20$self_db) # (n=248)
table(df_cardia_20$self_lab_dm) #(n=328)


#DM and TB
table(df_cardia_20$tb_history, df_cardia_20$self_db) # (n=12)
table(df_cardia_20$tb_history, df_cardia_20$self_lab_dm) # (n=17)
```

New DM and TB variable

```r
# For CARDIA Y00
df_cardia_00 <- df_cardia_00 %>%
  mutate(
    tb_dm = case_when(
      self_lab_dm == "Diabetes" & tb_history == "Yes" ~ 1,  # both DM and TB
      TRUE ~ 0                                               # everything else
    )
  )

# For CARDIA Y10
df_cardia_10 <- df_cardia_10 %>%
  mutate(
    tb_dm = case_when(
      self_lab_dm == "Diabetes" & tb_history == "Yes" ~ 1,
      TRUE ~ 0
    )
  )

# For CARDIA Y15
df_cardia_15 <- df_cardia_15 %>%
  mutate(
    tb_dm = case_when(
      self_lab_dm == "Diabetes" & tb_history == "Yes" ~ 1,
      TRUE ~ 0
    )
  )

# For CARDIA Y20
df_cardia_20 <- df_cardia_20 %>%
  mutate(
    tb_dm = case_when(
      self_lab_dm == "Diabetes" & tb_history == "Yes" ~ 1,
      TRUE ~ 0
    )
  )


# Labeling both datasets
label_tb_dm <- function(tb_dm) {
  tb_dm %>%
    mutate(
```

```
      # diabetes & tuberculosis
      tb_dm = factor(tb_dm,
                     levels = c(0, 1),
                     labels = c("No", "Yes"))
    )
}

df_cardia_00 <- label_tb_dm(df_cardia_00)
df_cardia_10 <- label_tb_dm(df_cardia_10)
df_cardia_15 <- label_tb_dm(df_cardia_15)
df_cardia_20 <- label_tb_dm(df_cardia_20)

#Checking
table(df_cardia_00$tb_dm)
table(df_cardia_10$tb_dm)
table(df_cardia_15$tb_dm)
table(df_cardia_20$tb_dm)
```

Preliminary results:

CARDIA Y00

```
library(gtsummary)

# Select the variables you want to include
vars_to_summarize <- c(
  "age", "sex", "race", "edu_adult", "self_db", "age_told_db", "drug_insu_db", "drug_pill_db",
  "self_lab_dm", "family_hx_db",
  "tb_history", "age_told_tb", "tb_treatment", "fasting_minutes", "fglu", "hba1c", "total_cholesterol",

# Create table1
CARDIA_Y10_table <- df_cardia_10 %>%
  select(all_of(vars_to_summarize), tb_dm) %>%  # include the grouping variable
  tbl_summary(
    by = tb_dm,  # group by TB+DM
    type = list(
      age ~ "continuous",
      age_told_db ~ "continuous",
      age_told_tb ~ "continuous",
      fglu ~ "continuous",
      hba1c ~ "continuous",
      fasting_minutes ~ "continuous",
      total_cholesterol ~ "continuous",
      ldl ~ "continuous",
      weight ~ "continuous",
      height ~ "continuous",
      bmi ~ "continuous",
      waist ~ "continuous",
      hip ~ "continuous",
      blood_pressure_sys ~ "continuous",
      blood_pressure_diast ~ "continuous"
```

```
    ),
    statistic = list(
      all_continuous() ~ "{mean} ({sd})",
      all_categorical() ~ "{n} / {N} ({p}%)"
    ),
    label = var_label(df_cardia_10),
    missing = "no"
  )

CARDIA_Y10_table <- CARDIA_Y10_table %>%
  modify_header(label ~ "**Variables**") %>%
  modify_spanning_header(c(stat_1, stat_2) ~ "**Summary Statistics (CARDIA Y10)**")

CARDIA_Y10_table
```

Preliminary results:

CARDIA Y15

```
# checking
table(df_cardia_15$tb_dm) # n=2
```

```
##
##   No  Yes
## 4546    2
```

```
# Due to inconsistency recording in the following participants in CARDIA studies, we are recording the
# 20575960
# 58702204
# 64049714

# Review inconsistencies
df_cardia_15 %>%
  filter(id %in% c(
    20575960,
    58702204,
    69644001
  )) %>%
  select(
    id,
    id_study,
    tb_history,
    self_db,
    self_lab_dm,
    tb_dm
  ) %>%
  arrange(id, id_study)
```

```
##         id   id_study tb_history self_db self_lab_dm tb_dm
```

```
## 1 20575960 CARDIA_Y15         No     Yes    Diabetes    No
## 2 58702204 CARDIA_Y15        Yes      No No Diabetes    No
## 3 69644001 CARDIA_Y15       <NA>    <NA>        <NA>    No
```

```r
# Recode
df_cardia_15 <- df_cardia_15 %>%
  mutate(
    # ID 20575960
    tb_history = if_else(
      id == 20575960,
      "Yes",
      tb_history
    ),

    tb_dm = if_else(
      id == 20575960,
      "Yes",
      tb_dm
    ),

    # ID 58702204
    self_db = if_else(
      id == 58702204,
      "Yes",
      self_db
    ),

    tb_dm = if_else(
      id == 58702204,
      "Yes",
      tb_dm
    ),

    # ID 69644001
    tb_history = if_else(
      id == 69644001,
      "Yes",
      tb_history
    ),

    self_db = if_else(
      id == 69644001,
      "Yes",
      self_db
    ),

    tb_dm = if_else(
      id == 69644001,
      "Yes",
      tb_dm
    )
  )

# Verify
```

```r
df_cardia_15 %>%
  filter(id %in% c(20575960, 58702204, 69644001)) %>%
  select(
    id,
    id_study,
    tb_history,
    self_db,
    self_lab_dm,
    tb_dm
  )
```

```
##         id   id_study tb_history self_db self_lab_dm tb_dm
## 1 20575960 CARDIA_Y15        Yes     Yes    Diabetes   Yes
## 2 58702204 CARDIA_Y15        Yes     Yes No Diabetes   Yes
## 3 69644001 CARDIA_Y15        Yes     Yes        <NA>   Yes
```

```r
# checking
table(df_cardia_15$tb_dm) # n=2 => 5
```

```
##
##   No  Yes
## 4542    5
```

```r
# re assigning labels
var_label(df_cardia_15) <- cardia_var_labels
```

```r
library(gtsummary)

CARDIA_Y15_table <- df_cardia_15 %>%
  select(all_of(vars_to_summarize), tb_dm) %>%  # include the grouping variable
  tbl_summary(
    by = tb_dm,  # group by TB+DM
    type = list(
      age ~ "continuous",
      age_told_db ~ "continuous",
      age_told_tb ~ "continuous",
      fglu ~ "continuous",
      hba1c ~ "continuous",
      fasting_minutes ~ "continuous",
      total_cholesterol ~ "continuous",
      ldl ~ "continuous",
      weight ~ "continuous",
      height ~ "continuous",
      bmi ~ "continuous",
      waist ~ "continuous",
      hip ~ "continuous",
      blood_pressure_sys ~ "continuous",
      blood_pressure_diast ~ "continuous"
    ),
    statistic = list(
      all_continuous() ~ "{mean} ({sd})",
      all_categorical() ~ "{n} / {N} ({p}%)"
```

```
    ),
    label = var_label(df_cardia_15)[vars_to_summarize],
    missing = "no"
  )

CARDIA_Y15_table <- CARDIA_Y15_table %>%
  modify_header(label ~ "**Variables**") %>%
  modify_spanning_header(c(stat_1, stat_2) ~ "**Summary Statistics (CARDIA Y15)**")

CARDIA_Y15_table
```

Preliminary results:

CARDIA Y20

```
# checking
table(df_cardia_20$tb_dm) # n=17
```

```
##
##   No  Yes
## 3077   17
```

```
# Due to inconsistency recording in the following participants in CARDIA studies, we are recording the
# 58702204
# 64049714

# Review inconsistencies
df_cardia_20 %>%
  filter(id %in% c(
    58702204,
    64049714
  )) %>%
  select(
    id,
    id_study,
    tb_history,
    self_db,
    self_lab_dm,
    tb_dm
  ) %>%
  arrange(id, id_study)
```

```
##         id   id_study tb_history self_db self_lab_dm tb_dm
## 1 58702204 CARDIA_Y20         No      No No Diabetes    No
## 2 64049714 CARDIA_Y20         No     Yes    Diabetes    No
```

```
# Recode
df_cardia_20 <- df_cardia_20 %>%
```

```r
  mutate(
    # ID 58702204
    tb_history = if_else(
      id == 58702204,
      "Yes",
      tb_history
    ),
      self_db = if_else(
      id == 58702204,
      "Yes",
      self_db
    ),

    tb_dm = if_else(
      id == 58702204,
      "Yes",
      tb_dm
    ),

    # ID 64049714
    tb_history = if_else(
      id == 64049714,
      "Yes",
      tb_history
    ),

    tb_dm = if_else(
      id == 64049714,
      "Yes",
      tb_dm
    )
  )

# Verify
df_cardia_20 %>%
  filter(id %in% c(58702204, 64049714)) %>%
  select(
    id,
    id_study,
    tb_history,
    self_db,
    self_lab_dm,
    tb_dm
  )
```

```
##         id   id_study tb_history self_db self_lab_dm tb_dm
## 1 58702204 CARDIA_Y20        Yes     Yes No Diabetes   Yes
## 2 64049714 CARDIA_Y20        Yes     Yes    Diabetes   Yes
```

```r
# checking
table(df_cardia_20$tb_dm) # n=17 => 19
```

```
##
```

```
##   No  Yes
## 3075   19
```

```r
# re assigning labels
var_label(df_cardia_20) <- cardia_var_labels
```

```r
library(gtsummary)

CARDIA_Y20_table <- df_cardia_20 %>%
  select(all_of(vars_to_summarize), tb_dm) %>%  # include the grouping variable
  tbl_summary(
    by = tb_dm,  # group by TB+DM
    type = list(
      age ~ "continuous",
      age_told_db ~ "continuous",
      age_told_tb ~ "continuous",
      fglu ~ "continuous",
      hba1c ~ "continuous",
      fasting_minutes ~ "continuous",
      total_cholesterol ~ "continuous",
      ldl ~ "continuous",
      weight ~ "continuous",
      height ~ "continuous",
      bmi ~ "continuous",
      waist ~ "continuous",
      hip ~ "continuous",
      blood_pressure_sys ~ "continuous",
      blood_pressure_diast ~ "continuous"
    ),
    statistic = list(
      all_continuous() ~ "{mean} ({sd})",
      all_categorical() ~ "{n} / {N} ({p}%)"
    ),
    label = var_label(df_cardia_15),
    missing = "no"
  )

CARDIA_Y20_table <- CARDIA_Y20_table %>%
  modify_header(label ~ "**Variables**") %>%
  modify_spanning_header(c(stat_1, stat_2) ~ "**Summary Statistics (CARDIA Y20)**")


CARDIA_Y20_table
```

CARDIA Y20 - manuscript

```r
# checking
table(df_cardia_20$tb_dm) # n=19

# "Clean" dataset
df_cardia_20 <- df_cardia_20 %>%
  filter(tb_dm == "Yes")          # n=19
```

```r
library(gtsummary)

CARDIA_Y20_table_manuscript <- df_cardia_20 %>%
  select(all_of(vars_to_summarize)) %>%   # remove tb_dm from the selection
  tbl_summary(
    type = list(
      age ~ "continuous",
      age_told_db ~ "continuous",
      age_told_tb ~ "continuous",
      fglu ~ "continuous",
      hba1c ~ "continuous",
      fasting_minutes ~ "continuous",
      total_cholesterol ~ "continuous",
      ldl ~ "continuous",
      weight ~ "continuous",
      height ~ "continuous",
      bmi ~ "continuous",
      waist ~ "continuous",
      hip ~ "continuous",
      blood_pressure_sys ~ "continuous",
      blood_pressure_diast ~ "continuous"
    ),
    statistic = list(
      all_continuous() ~ "{mean} ({sd})",
      all_categorical() ~ "{n} / {N} ({p}%)"
    ),
    label = var_label(df_cardia_20),
    missing = "no"
  ) %>%
  modify_header(label ~ "**Variables**") %>%
  modify_spanning_header(everything() ~ "**Summary Statistics (CARDIA Y20)**")


CARDIA_Y20_table_manuscript
```

Variables	Summary Statistics (CARDIA Y10)	
	No N = 4,539[1]	**Yes** N = 4[1]
Age	34.9 (3.6)	35.8 (2.1)
Sex		
Men	2,093 / 4,539 (46%)	0 / 4 (0%)
Women	2,446 / 4,539 (54%)	4 / 4 (100%)
Race		
Non-Hispanic Black	2,341 / 4,539 (52%)	4 / 4 (100%)
Non-Hispanic White	2,198 / 4,539 (48%)	0 / 4 (0%)
Mexican/Other Hispanic	0 / 4,539 (0%)	0 / 4 (0%)
Other/Multiracial	0 / 4,539 (0%)	0 / 4 (0%)
Missing	0 / 4,539 (0%)	0 / 4 (0%)
Education Level		
< High School	0 / 4,539 (0%)	0 / 4 (0%)
High School Grad	0 / 4,539 (0%)	0 / 4 (0%)
Some College/AA	0 / 4,539 (0%)	0 / 4 (0%)
College+	0 / 4,539 (0%)	0 / 4 (0%)
Refused/DK	4,539 / 4,539 (100%)	4 / 4 (100%)
Self-reported diabetes		
Yes	152 / 3,435 (4.4%)	4 / 4 (100%)
No	3,272 / 3,435 (95%)	0 / 4 (0%)
Refused/DK	11 / 3,435 (0.3%)	0 / 4 (0%)
Age told DM	27.9 (3.5)	27.8 (4.2)
Diabetes Mellitus on insulin		
Yes	0 / 0 (NA%)	0 / 0 (NA%)
No	0 / 0 (NA%)	0 / 0 (NA%)
Refused/DK	0 / 0 (NA%)	0 / 0 (NA%)
Diabetes Mellitus on Oral med		
Yes	0 / 0 (NA%)	0 / 0 (NA%)
No	0 / 0 (NA%)	0 / 0 (NA%)
Refused/DK	0 / 0 (NA%)	0 / 0 (NA%)
DM self-reported and by lab def		
Diabetes	183 / 3,441 (5.3%)	4 / 4 (100%)
No Diabetes	3,258 / 3,441 (95%)	0 / 4 (0%)
Family history of DM	0 / 0 (NA%)	0 / 0 (NA%)
TB History ever		
Yes	76 / 2,600 (2.9%)	4 / 4 (100%)
No	2,522 / 2,600 (97%)	0 / 4 (0%)
Refused/DK	2 / 2,600 (<0.1%)	0 / 4 (0%)
Age told TB	NA (NA)	NA (NA)
TB treatment ever		
Yes	0 / 0 (NA%)	0 / 0 (NA%)
No	0 / 0 (NA%)	0 / 0 (NA%)
Refused/DK	0 / 0 (NA%)	0 / 0 (NA%)
Fasting (minutes)	1,139 (9,732)	11,648 (21,592)
Fasting Glucose (mg/dl)	88 (20)	136 (83)
HbA1c (%)	NA (NA)	NA (NA)
Total cholesterol (mg/dL)	26 NA (NA)	NA (NA)
LDL (mg/dL)	109 (32)	103 (33)
Weight (lbs)	80 (20)	91 (23)
Height (cm)	171 (9)	162 (4)

Variables	Summary Statistics (CARDIA Y15)	
	No N = 4,542[1]	**Yes** N = 5[1]
Age	24.7 (3.6)	25.6 (2.3)
Sex		
Men	2,093 / 4,538 (46%)	0 / 5 (0%)
Women	2,445 / 4,538 (54%)	5 / 5 (100%)
Race		
Non-Hispanic Black	2,340 / 4,538 (52%)	5 / 5 (100%)
Non-Hispanic White	2,198 / 4,538 (48%)	0 / 5 (0%)
Mexican/Other Hispanic	0 / 4,538 (0%)	0 / 5 (0%)
Other/Multiracial	0 / 4,538 (0%)	0 / 5 (0%)
Missing	0 / 4,538 (0%)	0 / 5 (0%)
Education Level		
< High School	0 / 4,542 (0%)	0 / 5 (0%)
High School Grad	1,266 / 4,542 (28%)	1 / 5 (20%)
Some College/AA	1,220 / 4,542 (27%)	1 / 5 (20%)
College+	530 / 4,542 (12%)	0 / 5 (0%)
Refused/DK	1,526 / 4,542 (34%)	3 / 5 (60%)
Self-reported diabetes		
No	2,979 / 3,185 (94%)	0 / 5 (0%)
Refused/DK	19 / 3,185 (0.6%)	0 / 5 (0%)
Yes	187 / 3,185 (5.9%)	5 / 5 (100%)
Age told DM	32.6 (3.9)	31.0 (4.4)
Diabetes Mellitus on insulin		
Yes	0 / 0 (NA%)	0 / 0 (NA%)
No	0 / 0 (NA%)	0 / 0 (NA%)
Refused/DK	0 / 0 (NA%)	0 / 0 (NA%)
Diabetes Mellitus on Oral med		
Yes	0 / 0 (NA%)	0 / 0 (NA%)
No	0 / 0 (NA%)	0 / 0 (NA%)
Refused/DK	0 / 0 (NA%)	0 / 0 (NA%)
DM self-reported and by lab def		
Diabetes	210 / 3,188 (6.6%)	3 / 4 (75%)
No Diabetes	2,978 / 3,188 (93%)	1 / 4 (25%)
Family history of DM	0 / 0 (NA%)	0 / 0 (NA%)
TB History ever		
No	3,078 / 3,190 (96%)	0 / 5 (0%)
Refused/DK	3 / 3,190 (<0.1%)	0 / 5 (0%)
Yes	109 / 3,190 (3.4%)	5 / 5 (100%)
Age told TB	27.0000 (0.0000)	27.0000 (0.0000)
TB treatment ever		
Yes	0 / 0 (NA%)	0 / 0 (NA%)
No	0 / 0 (NA%)	0 / 0 (NA%)
Refused/DK	0 / 0 (NA%)	0 / 0 (NA%)
Fasting (minutes)	855 (1,788)	855 (100)
Fasting Glucose (mg/dl)	87 (21)	168 (80)
HbA1c (%)	NA (NA)	NA (NA)
Total cholesterol (mg/dL)	27 NA (NA)	NA (NA)
LDL (mg/dL)	113 (32)	82 (25)
Weight (lbs)	84 (20)	79 (23)
Height (cm)	171 (9)	162 (4)

Variables	Summary Statistics (CARDIA Y20)	
	No N = 3,075[1]	**Yes** N = 19[1]
Age	25.0 (3.6)	24.8 (3.7)
Sex		
Men	1,346 / 3,059 (44%)	4 / 19 (21%)
Women	1,713 / 3,059 (56%)	15 / 19 (79%)
Race		
Non-Hispanic Black	1,418 / 3,059 (46%)	18 / 19 (95%)
Non-Hispanic White	1,632 / 3,059 (53%)	1 / 19 (5.3%)
Mexican/Other Hispanic	9 / 3,059 (0.3%)	0 / 19 (0%)
Other/Multiracial	0 / 3,059 (0%)	0 / 19 (0%)
Missing	0 / 3,059 (0%)	0 / 19 (0%)
Education Level		
< High School	0 / 3,075 (0%)	0 / 19 (0%)
High School Grad	1,162 / 3,075 (38%)	8 / 19 (42%)
Some College/AA	1,182 / 3,075 (38%)	9 / 19 (47%)
College+	562 / 3,075 (18%)	1 / 19 (5.3%)
Refused/DK	169 / 3,075 (5.5%)	1 / 19 (5.3%)
Self-reported diabetes		
No	2,793 / 3,058 (91%)	5 / 19 (26%)
Refused/DK	30 / 3,058 (1.0%)	0 / 19 (0%)
Yes	235 / 3,058 (7.7%)	14 / 19 (74%)
Age told DM	37.2 (4.6)	38.2 (5.5)
Diabetes Mellitus on insulin		
Yes	0 / 0 (NA%)	0 / 0 (NA%)
No	0 / 0 (NA%)	0 / 0 (NA%)
Refused/DK	0 / 0 (NA%)	0 / 0 (NA%)
Diabetes Mellitus on Oral med		
Yes	174 / 224 (78%)	11 / 13 (85%)
No	50 / 224 (22%)	2 / 13 (15%)
Refused/DK	0 / 224 (0%)	0 / 13 (0%)
DM self-reported and by lab def		
Diabetes	310 / 3,059 (10%)	18 / 19 (95%)
No Diabetes	2,749 / 3,059 (90%)	1 / 19 (5.3%)
Family history of DM	0 / 0 (NA%)	0 / 0 (NA%)
TB History ever		
No	2,945 / 3,058 (96%)	0 / 19 (0%)
Refused/DK	5 / 3,058 (0.2%)	0 / 19 (0%)
Yes	108 / 3,058 (3.5%)	19 / 19 (100%)
Age told TB	31.09 (3.28)	29.58 (3.20)
TB treatment ever		
Yes	0 / 0 (NA%)	0 / 0 (NA%)
No	0 / 0 (NA%)	0 / 0 (NA%)
Refused/DK	0 / 0 (NA%)	0 / 0 (NA%)
Fasting (minutes)	4,408 (57,218)	850 (171)
Fasting Glucose (mg/dl)	100 (25)	143 (64)
HbA1c (%)	NA (NA)	NA (NA)
Total cholesterol (mg/dL)	28 NA (NA)	NA (NA)
LDL (mg/dL)	110 (32)	89 (25)
Weight (lbs)	86 (20)	90 (23)
Height (cm)	171 (9)	164 (7)

Summary Statistics (CARDIA Y20)	
Variables	**N = 19**[1]
Age	24.8 (3.7)
Sex	
Men	4 / 19 (21%)
Women	15 / 19 (79%)
Race	
Non-Hispanic Black	18 / 19 (95%)
Non-Hispanic White	1 / 19 (5.3%)
Mexican/Other Hispanic	0 / 19 (0%)
Other/Multiracial	0 / 19 (0%)
Missing	0 / 19 (0%)
Education Level	
< High School	0 / 19 (0%)
High School Grad	8 / 19 (42%)
Some College/AA	9 / 19 (47%)
College+	1 / 19 (5.3%)
Refused/DK	1 / 19 (5.3%)
Self-reported diabetes	14 / 19 (74%)
Age told DM	38.2 (5.5)
Diabetes Mellitus on insulin	
Yes	0 / 0 (NA%)
No	0 / 0 (NA%)
Refused/DK	0 / 0 (NA%)
Diabetes Mellitus on Oral med	
Yes	11 / 13 (85%)
No	2 / 13 (15%)
Refused/DK	0 / 13 (0%)
DM self-reported and by lab def	
Diabetes	18 / 19 (95%)
No Diabetes	1 / 19 (5.3%)
Family history of DM	0 / 0 (NA%)
TB History ever	
Yes	19 / 19 (100%)
Age told TB	29.58 (3.20)
TB treatment ever	
Yes	0 / 0 (NA%)
No	0 / 0 (NA%)
Refused/DK	0 / 0 (NA%)
Fasting (minutes)	850 (171)
Fasting Glucose (mg/dl)	143 (64)
HbA1c (%)	NA (NA)
Total cholesterol (mg/dL)	NA (NA)
LDL (mg/dL)	89 (25)
Weight (lbs)	90 (23)
Height (cm)	164 (7)
BMI (kg/m2)	33 (9)
Waist (cm)	98 (15)
Hip (cm)	NA (NA)
Systolic blood pressure (mmHg)	121 (15)
Diastolic blood pressure (mmHg)	76 (11)